

Vocab Learning Final Report

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1. Introduction

Vocabulary is a foundational element for language mastery, underpinning comprehension, communication, and fluency. As Chinese international students who have navigated the complexities of learning English as a second language, we recognize the critical need for vocabulary-building tools that are both accessible and effective. Many existing vocabulary apps lack features tailored to the specific requirements of students aiming to excel in standardized English proficiency exams such as the TOEFL. Additionally, these apps often fall short in providing adaptable orientations and compatibility with larger screens like tablets and foldable devices, which are increasingly popular among students.

Our app seeks to bridge this gap by offering a responsive and engaging vocabulary-learning experience specifically designed for Chinese international students with a basic English background. The app is meticulously curated to include all the vocabulary necessary to pass the TOEFL exam, ensuring that users can enhance their English skills in a targeted and efficient manner. By adapting seamlessly across various devices and providing a user-friendly interface, our app makes language learning more convenient and enjoyable, ultimately helping students achieve their academic and professional goals.

2. Targeted Customers

Our Vocabulary Learning App is meticulously crafted to cater to the unique needs of Chinese international students who are preparing to take the Test of English as a Foreign Language (TOEFL). These students typically possess a foundational understanding of English and are striving to improve their language skills to meet the academic requirements of universities and colleges abroad. The app targets the following customer segments:

a) Chinese International Students Preparing for TOEFL

- **Demographics:**
 - **Age:** Primarily aged 18-30.
 - **Education Level:** High school graduates, undergraduate students, and postgraduate applicants.

- **Geographic Location:** Predominantly located in China, including major cities such as Beijing, Shanghai, Guangzhou, and others.
- **Psychographics:**
 - **Goals:** Achieve a high score on the TOEFL exam to secure admission to reputable international universities.
 - **Challenges:** Struggle with expanding their English vocabulary, understanding nuanced meanings, and applying words effectively in academic contexts.
 - **Behavior:** Regularly engage in language learning, seek efficient and effective study tools, and prefer resources that align with their exam preparation needs.

b) English Language Enthusiasts

- **Demographics:**
 - **Age:** All adult age groups, particularly those between 18-35.
 - **Education Level:** College students, professionals, and lifelong learners.
 - **Geographic Location:** Global, with a focus on regions with significant Chinese-speaking populations.
- **Psychographics:**
 - **Goals:** Enhance English vocabulary for personal development, career advancement, or academic pursuits.
 - **Challenges:** Find comprehensive and structured vocabulary resources that cater to advanced learning without being overwhelming.
 - **Behavior:** Actively seek out language improvement tools, value structured learning paths, and appreciate user-friendly, intuitive interfaces.

c) Educational Institutions and Language Centers

- **Demographics:**
 - **Type:** Universities, language schools, and educational centers in China and other countries with Chinese student populations.
 - **Psychographics:**
 - **Goals:** Provide effective TOEFL preparation resources to their students.
 - **Challenges:** Need reliable, evidence-based tools that can be integrated into their curriculum and track student progress.
 - **Behavior:** Partner with app developers to incorporate technology into their teaching methods, value comprehensive and up-to-date content, and require tools that offer analytics and reporting features.
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3. Functionality

Mo_Li Vocab Learning App is a vocabulary learning app that users can learn advanced English vocabulary, set up learning goal, take quizzes, search word in English and Chinese, save favorite words, review unmastered words, async learning progress with accounts, enable dark mode, and receive push notifications as a reminder. The app is fully responsive and color-schemed.

Our unique app feature is that users can learn the words both in **landscape** and portrait modes, while all the other vocabulary learning apps are only available in portrait mode. People usually turn the phone into landscape when they want to dig into some contents to read carefully. The users can now turn the phone horizontally and focus more on their screen to gain a better learning experience. Imagine that you learn the English words in a gaming-like gesture while holding the phone, isn't that cool?

Another cornerstone of the app is the **learning goal setup** feature, allowing users to define specific number of words to learn for their individual learning pace and targets. Whether aiming to learn a set number of new words each week or achieving proficiency in particular vocabulary categories, this functionality ensures that users have clear milestones to move towards.

To enhance learning and assess comprehension, the app incorporates interactive **quizzes** designed to test users' knowledge of the vocabulary they have studied. These quizzes offer immediate typing feedback, identifying words that require further review or they are being mastered. By engaging in regular quizzes, users can solidify their memorizations of the vocabulary, which is essential for getting good scores in exams like the TOEFL.

The app's **search functionality** is another pivotal feature, enabling users to look up words in both English and Chinese. This bilingual search capability allows learners to explore vocabulary within their current learning word book. Users can efficiently access the words with translations they need to fulfill their instant goal, such as when they encounter unfamiliar words during reading.

Enhancing the personalized learning experience, users can **save favorite words**, creating a list of vocabulary that is tailored to the user. This favorite feature allows learners to revisit and review selected words at their convenience, facilitating targeted study sessions and ensuring that important vocabulary is readily accessible for continuous reinforcement.

For words that remain **unmastered**, the app offers a dedicated **review** mechanism. This functionality identifies and highlights vocabulary that users have struggled with, prompting focused quizzes to allow users to memorize these words multiple times. By systematically revisiting challenging words, the app promotes comprehensive retention and ensures that users build a robust and well-rounded vocabulary foundation.

As we know the importance of **asynchronous learning progress**, the app supports account-based data management. Users can create and log into their accounts, ensuring that their learning progress is securely stored and synchronized across multiple devices. This data integration allows learners to continue their studies uninterrupted.

To enhance visual comfort, the app includes a **dark mode** option. This feature allows users to switch to a dark tinted color scheme and provide a visually soothing interface in low-light environments.

Finally, the app is equipped with **push notifications** to serve as timely reminders, encouraging users to adhere to their study schedules and maintain consistent engagement with their learning goals. These notifications can prompt users to complete daily lessons, take quizzes, or review challenging words. Specifically, the notifications will aware users periodically about their unmastered words to help users to learn difficult words in their spare time or in some short time intervals, such as taking a bus.

4. Technical implementation(backend and frontend)

a) Authentication:

The app leverages **Firebase Authentication** to use email/passwor authentication method providing a robust and user-friendly authentication mechanism.

b) Real-time Data Sync and WorkManager

To maintain consistent and up-to-date user data across multiple devices, our app integrates **Firebase Firestore** for real-time data synchronization and **WorkManager** for scheduling periodic background tasks. Additionally, we utilize **DataStore** for synchronizing user app settings, ensuring a seamless and personalized user experience.

Firebase Firestore:

Upon user authentication using Firebase Authentication, we initialize Firestore and set up a remote database specific to the authenticated user. We store the user's learning progress, including the list of words being learned and their statuses, in Firestore under the user's document. This ensures that any changes made on one device are instantly reflected across all devices where the user is logged in. To handle potential conflicts between local and remote data, especially when users make updates from different devices simultaneously, we implement

a conflict resolution strategy based on the `lastUpdated` timestamp. This ensures that the most recent changes are preserved.

Scheduling Periodic Data Synchronization with WorkManager:

We create a `PeriodicWorkRequest` that triggers the `DataSyncWorker` every hour, ensuring timely synchronization of user data with Firestore. The `DataSyncWorker` class extends `CoroutineWorker` to perform the synchronization tasks asynchronously, leveraging Kotlin Coroutines for efficient background processing.

Synchronizing User App Settings with DataStore:

In addition to synchronizing core user data, we employ **DataStore** for managing and syncing user-specific app settings. We set up DataStore to handle user settings such as preferred themes (light or dark mode), notification preferences, and other customizable options. To ensure that user settings are consistent across all devices, we synchronize DataStore with Firestore. When a user updates their settings on one device, the changes are reflected on all other devices upon synchronization. Similar to data synchronization, we schedule periodic synchronization of user settings using WorkManager to ensure settings remain consistent across devices.

c) Data Persistence

To ensure the app remains functional even without an active internet connection, we utilize **Room**. We define the Room database with several entities representing the vocabulary, user progress, and quiz records. we also created the DAOs (Data Access Objects) to facilitate seamless interaction with the database.

Data Caching Strategies:

To further optimize performance and reduce unnecessary network usage, the app implements strategic caching mechanisms. These strategies focus on caching essential data upon specific user actions, such as selecting a set of words to learn or initiating a quiz.

Caching Strategy Overview:

Trigger Points for Caching:

- **Word Selection:** When a user selects the number of words they wish to learn (e.g., selecting 10 words), the app caches the corresponding vocabulary data and associated audio files.
- **Quiz Initiation:** Before starting a quiz, the app caches all necessary data for the words included in the quiz, ensuring that all resources are readily available.

Cached Data:

- Audio Files for Pronunciation:

- Word phonetic and example sentences

d) Search and Filtering:

To enhance user experience and facilitate efficient navigation through the vocabulary list, the app incorporates a **search bar** accompanied by **filtering options**. This enables users to swiftly locate specific words and tailor their learning sessions based on various criteria. A search bar is integrated into the UI using Jetpack Compose's `TextField`. It captures user input and updates the search query state. The `ViewModel` maintains a `FilterState` data class that holds the current filter criteria, such as word status ("unseen", "learning", "mastered") and favorite status. Filtering options are presented using dropdown menus or toggle buttons, allowing users to select their desired filters. The `ViewModel` utilizes `StateFlow` to manage and emit the current search query and filter state. This ensures that the UI reacts reactively to changes in these states.

E) Networking and API Integration

The app integrates the **Free Dictionary API** to fetch comprehensive information about vocabulary words, including pronunciation audio files, phonetic transcriptions, and example sentences. This integration enriches the learning experience by providing users with detailed word data essential for mastering vocabulary.

The app utilizes **Retrofit** in combination with **Kotlin Coroutines** to handle network requests asynchronously, ensuring efficient and non-blocking data fetching. By declaring the function as `suspend`, the app seamlessly integrates with Kotlin Coroutines, allowing for non-blocking network calls. The app also utilizes Moshi as the JSON converter, enabling efficient parsing of JSON responses into Kotlin data classes. Each data class is annotated with `@JsonClass(generateAdapter = true)` to enable Moshi to generate the necessary adapters for JSON parsing. The models mirror the structure of the Free Dictionary API responses, ensuring accurate and efficient data parsing. To handle errors, we use Try-Catch Blocks, Encapsulates network operations within try-catch blocks to capture and handle exceptions such as network timeouts, unreachable servers, or malformed responses, preventing the app from crashing due to unhandled exceptions and allows for graceful degradation of functionality.

In developing the frontend of the Vocabulary Learning App, a primary focus was placed on creating an intuitive and engaging user interface that accommodate diverse learning needs. Based on Jetpack Compose, the design process emphasized modularity, responsiveness, and seamless user interactions. The frontend designs include all main screens, learning goal selection, word learning sections both in landscape and portrait modes, quiz sections, and

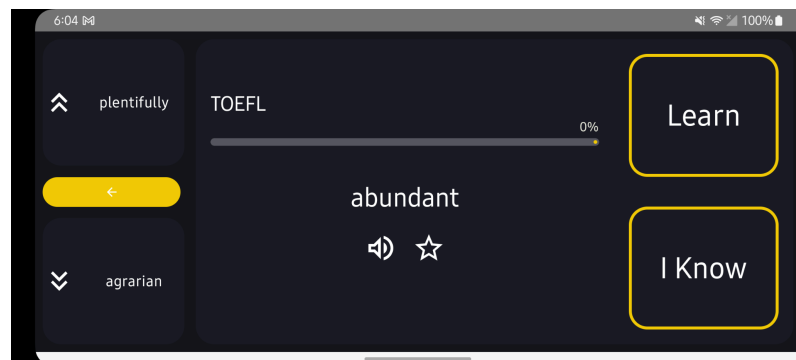
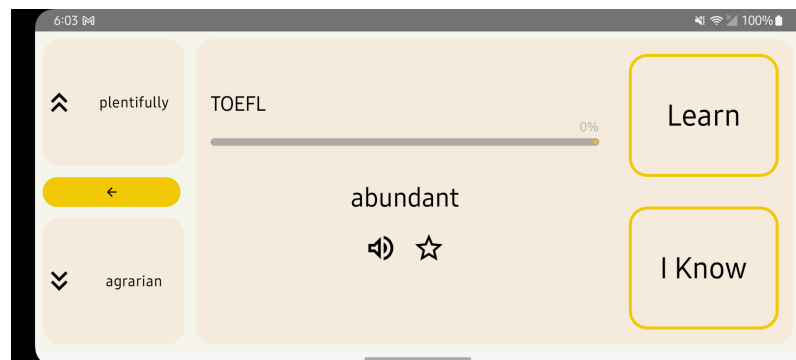
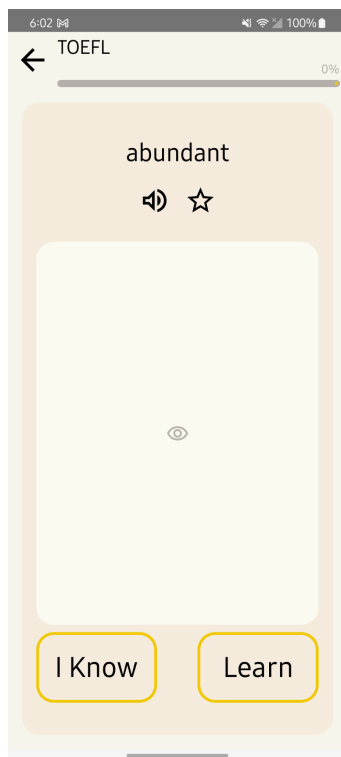
search screen designs. We used a consistent yellow color as our primary color scheme, as yellow is an energizing color that makes us happy and optimistic and can improve our memory.

What we designed:

We used tinted backgrounds under texts to highlight, as well as yellow-tinted borders for a modern looking. We utilized meaningful icons for better illustrations, and dynamic progress bar for learning progress tracking. We designed the dark and light color schemes and kept all the places where colors consistent for a better user experience.

Techniques we used:

We applied Material3, ViewModel, state management with StateFlow, Jetpack Navigation Compose, responsive layouts with LazyColumn and COnstraintLayout into our final project.



5. Challenges

One of the main challenges we faced was setting overly ambitious goals. Initially, we aimed to include numerous functionalities and target a broad market. However, we soon realized that trying to achieve all these objectives within the given timeframe and resources was not practical. This required us to reassess our priorities, scale down the scope of the project, and focus on delivering a streamlined and effective application.

Another challenge was that when building complex screen with many boxes, rows, columns, the arrangement of these UIs became very annoying that it took a lot of builds and runs to figure out one proper way to arrange the elements in a screen. For example, the code base for learning flash card or quiz section can have up to 800 lines of code, it acquired enough patience to handle the arrangement of the screens.

6. Features that were modified or omitted

In the original proposal, we planned to use AI to generate quizzes for users. However, after careful consideration, we opted to replace this functionality with pre-programmed quizzes to ensure greater reliability and accuracy. Additionally, support for smartwatches was not included in the final implementation. The app's scope was also refined, transitioning from a general language learning platform to a more focused TOEFL vocabulary learning app. Finally, we decided to maintain the interface in English, as it provides an immersive environment that is more conducive to learning the language effectively.

